Lecture 8: Legal texts as data

Empirical Methods for Law and Economics | EDLE

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Today's questions

How do we turn a constitution into a number?

How do we know the number measures what we want?

- By the end of today you will be able to:
  - describe the three steps that turn text into data;
  - build a dictionary measure from legal texts in Stata;
  - validate it: read what it counts, look for what it misses, compare it with other measures;
  - revise your draft referee comments.

DISCUSS De jure independence, coded by hand

- Voigt and Wulf measure the *de jure* independence of prosecutors with up to 21 items, coded by reading legal texts for 76 countries (Lecture 4).

Questions:

1. What does hand-coding cost, and what can go wrong?
2. Could a computer read the texts instead? What would it do well, and what badly?

ANSWERS De jure independence, coded by hand

1. Hand-coding costs expert time for every country and every change in the law. Two coders can read the same article differently, and a coder who expects a result may find it. The coding rules are hard for others to reproduce.

2. A computer applies the same rule to every text, at no cost, and anyone can rerun it. But it reads words, not meaning: "independent" in a provision on courts and "independent" in the name of a tribunal look the same to it.

- Today we build such a measure, then check it as carefully as a hand-coded one.

Text as data in three steps

1. Represent raw text as numbers: for each document, the presence or count of words or phrases.
2. Map those numbers into a measure of what we care about.
3. Use the measure in a descriptive or causal analysis.

- Each step involves choices. The first: **what is a document?** A constitution, an article, a sentence?
- The simplest way to do step 2 is a **dictionary**: a list of words chosen by the researcher. Dictionary methods "do not involve statistical inference at all" and are "by far the most common method" in the social sciences.

Gentzkow, M., Kelly, B. and Taddy, M. (2019), *Journal of Economic Literature* 57(3): 535–574, pp. 536 and 541.

Our corpus

- The constitutions of 25 European countries, in English, from Constitute Project (licence CC BY-NC 3.0): 25 PDF files.
- Most are translations, not the legal texts. Each includes amendments up to a different year (Italy: 2012).
- Two steps before we can count anything:
  1. **from PDF to text**: keep the text of the constitution, drop what the page adds for the reader;
  2. **from text to data**: one row per sentence or clause.

Elkins, Z., Ginsburg, T. and Melton, J., *Constitute: The World's Constitutions to Read, Search, and Compare*, constituteproject.org. Details in `text_analysis/LICENCE_AND_SOURCE.txt`.

From PDF to text: what is on the page



Page header: dropped (position on the page)

Headings (large font): start a new sentence

Body text: kept, split into sentences

Margin labels: dropped (small font)

- The program `pdftotext` returns every word with its position on the page and the size of its font. The rules on the next slide use both: font size identifies margin labels and headings, position identifies headers and footers.
- The cover page, the table of contents and the topic index are also dropped.

Italy's Constitution of 1947 with amendments through 2012, p. 4, constituteproject.org.

From PDF to text: the script

- `text_analysis/08_pdf_to_txt.py` (Python; it uses the program `pdftotext`). The core rules, slightly simplified:

```
# keep a word only if its font is not small (margin labels)
# and it is not at the top or bottom of the page (headers, footers)
ws = [w for w in words(page) if height(w) > 9.5 and 35 < top(w) < 750]

# a line in a large font is a heading: it closes the previous sentence
kind = "H" if height >= 13.5 else "B"

# a sentence ends with . ; or : followed by a capital letter, a digit,
# a bracket or a quotation mark
re.split(r'(?<=[.;:])\s+(?=[A-ZO-9(\["])', text)
```

- Every threshold is a choice made by looking at the PDFs. Change it, and the data change.
- Output, in `text_analysis/constitutions_txt/`: one file per constitution with one sentence or clause per line (e.g. `Italy_2012.txt`), and `constitutions_list.csv` with one row per constitution.

STATA Guided Stata 1: from text files to a data set

```
global text_dir "../../text_analysis/constitutions_txt"
import delimited using "$text_dir/constitutions_list.csv", ///
    varnames(1) encoding("utf-8") clear
save "$course_results/tables/08_constitutions_list.dta", replace
levelsof constitute_id, local(ids)
clear
tempfile corpus
save `corpus', emptyok
foreach id of local ids {
    import delimited text using "$text_dir/`id'.txt", ///
        delimiter(tab) varnames(nonames) bindquote(nobind) stripquote(no) ///
        stringcols(_all) encoding("utf-8") clear
    generate constitute_id = "`id'"
    generate sent_id = _n
    append using `corpus'
    save `corpus', replace
}
merge m:1 constitute_id using "$course_results/tables/08_constitutions_list.dta", nogenerate
sort constitute_id sent_id
order economy_code country constitute_id year_adopted year_text sent_id text
save "$course_data/constitutions_text.dta", replace
```

- Each line of a text file becomes a row; `foreach` repeats the import for the 25 files and `append` stacks them.
- Result: 18,853 rows, saved as `data/constitutions_text.dta`. The copy you received in `data/` is identical: if the loop fails, continue from Guided Stata 2.

STATA Guided Stata 2: the corpus

```
use "$course_data/constitutions_text.dta", clear
describe
generate words = wordcount(subinstr(text, char(34), "", .))
tabstat words, by(country) statistics(n sum)
```

Country	Sentences	Words
Sweden	2,190	59,285
Austria	1,795	49,362
	...	
Denmark	246	6,163
Latvia	219	4,962
All 25	18,853	466,841

- Sweden's text is 12 times longer than Latvia's. Any count of sentences must therefore be divided by a measure of length.

PREDICT Who writes about the independence of judges?

Before any code: name two constitutions that you expect to say a lot about the independence of judges, and two that you expect to say little. Why?

- Hint: when was each constitution written, and what did its authors want to protect the courts from?

A dictionary

- A sentence is about the **judiciary** if it contains one of: judge, judges, judicial, judiciary, court, courts, tribunal, tribunals.
- A sentence mentions **independence** if it contains a word starting with independen- or impartial-.
- A sentence is **flagged** if it does both.
- For each country, the measure is the **share** of judiciary sentences that are flagged. A share and not a count, because constitutions differ in length.

STATA Guided Stata 3: applying the dictionary

```
generate text_lc = ustrlower(text)
generate byte judiciary = ustrregexm(text_lc, ///
    "\b(judge|judges|judicial|judiciary|court|courts|tribunal|tribunals)\b")
generate byte indep = ustrregexm(text_lc, "\b(independen|impartial)")
generate byte jud_indep = judiciary & indep
tabulate judiciary jud_indep
```

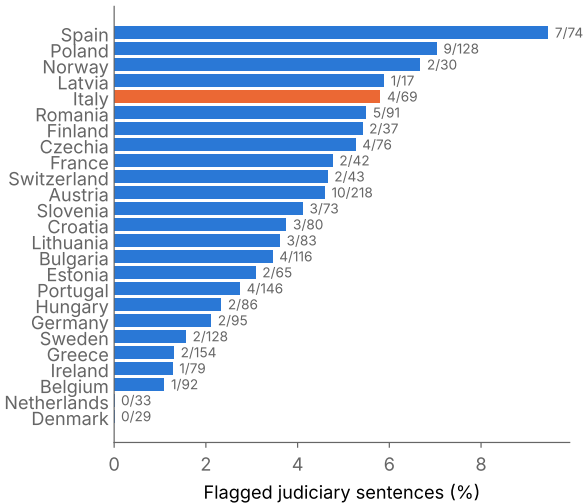
- `ustrregexm` returns 1 if the text matches a **regular expression**: `|` means "or", `\b` marks the edge of a word.
- Result: 2,084 sentences about the judiciary; 77 of them flagged.

STATA Guided Stata 4: one number per constitution

```
preserve
collapse (count) sentences=sent_id (sum) words judiciary jud_indep, ///
    by(economy_code country)
generate share = 100*jud_indep/judiciary
gsort -share
list country sentences words judiciary jud_indep share, noobs
graph hbar share, over(country, sort(1) descending)
save "$course_results/tables/08_constitutions_country.dta", replace
restore
```

- `collapse` turns 18,853 sentences into 25 rows, one per constitution: the number of sentences, and the sums of words and of the two 0/1 indicators.
- The country file is saved for Guided Stata 6; `restore` brings back the sentences.

The measure



- Next to each bar: flagged sentences / judiciary sentences.
- Spain and Poland have the highest shares; the Netherlands and Denmark have none.
- Italy: 4 of 69 judiciary sentences, 5.8%.

DISCUSS Reading the ranking

Questions:

1. Does Spain's constitution protect judicial independence more than any other?
2. The Netherlands and Denmark score zero. Do their judges lack independence?
3. Austria has 10 flagged sentences, more than any other country. What would you check first?

ANSWERS Reading the ranking

1. Not necessarily. The share counts sentences that use certain words, not the strength of the guarantees. A constitution can repeat “independent” often and protect judges weakly, or say it once in a provision that is hard to amend.

2. Not necessarily. A zero means that no sentence about courts uses these words. The guarantee may be written with other words, or rest on ordinary law and practice rather than on the constitution.

3. Read the ten sentences. When a country is far from the others, the first question is what was counted.

Validation

- Gentzkow, Kelly and Taddy recommend a manual audit: "cross-checking some subset of the fitted values against the coding a human would produce". "Often as few as twenty or thirty documents is enough", and the audit "is especially important for dictionary methods".
- The audit answers two questions:
  - **Precision**: of the sentences we flagged, what share is really about the independence of courts?
  - **Recall**: of the sentences really about it, what share did we flag?
- A third check: compare the measure with other measures of the same concept.

Gentzkow, Kelly and Taddy (2019), Section 3.5.2, p. 555.

STATA Guided Stata 5: read what you counted

```
set seed 20261030
generate double u = runiform()
gsort -jud_indep u
list country text in 1/20, noobs
```

- `runiform()` gives each sentence a random number; `gsort` puts the flagged sentences first, in random order. The first 20 rows are a random sample of the flagged sentences.
- Code each sentence 1 if it is about the independence or impartiality of courts that decide cases, or of their judges; 0 otherwise.

DISCUSS Six flagged sentences

1. Italy: "The Judiciary is a branch that is autonomous and independent of all other powers."
2. Austria: "Per 1. July 2008 the former independent Federal Asylum Tribunal becomes the Asylum Court."
3. Germany: "The Federal Court of Audit, whose members shall enjoy judicial independence, shall audit the account [...]"
4. France: "Statutes shall determine the rules concerning: civic rights [...]; freedom, diversity and the independence of the media; [...]"
5. Hungary: "Everyone shall have the right to have any charge against him or her [...] adjudicated [...] in a fair and public trial by an independent and impartial court established by an Act."
6. Croatia: "The Public Prosecution Service is an autonomous and independent judicial body [...]"

Correct or not? For each error, say which word misled the dictionary.

ANSWERS Six flagged sentences

1. Italy: correct.
2. Austria: wrong. “Independent” is part of the name of a tribunal, in a transitional provision.
3. Germany: wrong under our rule. “Court” is part of the name of the Federal Court of Audit, which checks public accounts and decides no cases; “judicial independence” describes the status of its members.
4. France: wrong. “Independence” refers to the media; “courts” appears later in the same long sentence.
5. Hungary: correct. A right to an independent and impartial court.
6. Croatia: wrong. The sentence is about prosecutors, although it calls them a “judicial body”.

- I read all 77 flagged sentences: 15 are wrong, so **precision is 81%**.
- Nine of the 15 errors are Austrian (seven on the asylum tribunal, two on independent administrative authorities). Austria has 1 correct sentence, not 10: the errors change its rank, and an average precision does not show it.

What the dictionary misses

- Denmark, section 64: “In the performance of their duties the judges shall be directed solely by the law.”
- Italy, Article 101: “Judges are subject only to the law.”
- France, Article 64: “Judges shall be irremovable from office.”

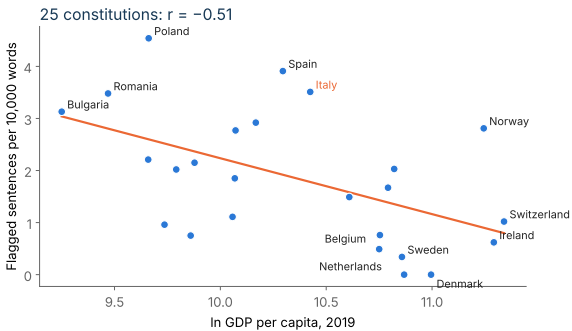
- These sentences protect the independence of judges with other words, and the dictionary misses them: **recall** is below 100%. How far below, we learn only by reading sentences that were not flagged.
- A second dictionary adds irremovab-, “subject only to”, “bound only by” and “solely by the law”: 86 flagged sentences instead of 77. Denmark goes from 0 to 1, Italy from 4 to 5, France from 2 to 3, Bulgaria from 4 to 7.
- Each new word can add errors too, so the second dictionary must be audited again.

STATA Guided Stata 6: compare with other measures

```
use "$course_results/tables/08_constitutions_country.dta", clear
merge 1:1 economy_code using "$course_data/justice_europe_2019.dta", ///
    keep(match) nogenerate
generate per10k = 10000*jud_indep/words
generate ln_gdppc = ln(gdp_pc_usd)
pwcorr share judicial_quality, sig obs
pwcorr per10k ln_gdppc, sig obs
```

- `keep(match)` keeps the 25 countries found in both files; `nogenerate` skips the `_merge` variable.
- The share and the Doing Business index of the quality of judicial processes: $r = 0.20$ ($p = 0.33$, 25 countries). The text measure is only weakly related to this index of how courts work.
- Flagged sentences per 10,000 words and ln GDP per capita: $r = -0.51$ ($p = 0.009$). With the share instead: $r = -0.22$ ($p = 0.30$).
- Neither variable measures judicial independence: these are weak checks. A proper one needs another measure of the same concept, for example a hand-coded index of de jure judicial independence.

DISCUSS Income and the text measure



Questions:

1. How would you explain the negative correlation with income?
2. Does it show that the text measure is invalid?
3. What does it remind you of in Voigt and Wulf?
4. The correlation is -0.51 with the per-word measure and -0.22 with the share. What should a paper report?

ANSWERS Income and the text measure

1. One hypothesis: constitutions written after an authoritarian regime (Spain 1978, Poland 1997) spell out the guarantees; older ones (the Netherlands, Denmark, Belgium) leave them to ordinary law and practice. Norway (1814, third) and Hungary (2011, near the bottom) do not fit: with 25 countries this is a hypothesis, not a result.
2. No. The measure describes what the constitution says, not how courts work. Both are legitimate objects of study, but a paper must say which one it needs.
3. If richer countries have courts that work better in practice, a text measure that falls with income resembles the gap in Voigt and Wulf: de jure and de facto independence of prosecutors are weakly and negatively correlated ($r = -0.21$, from Van Aaken et al. 2010).
4. Both, and why they differ. The per-word measure divides by the whole text, so it also falls when a constitution is long for reasons unrelated to courts; the share compares only sentences about courts. A result that holds with one measure only is weaker.

A referee's checklist for text-based measures

- **Documents**: which texts, which version, which language? Translations are not the legal text (the Swiss English version "has no legal force").
- **Unit**: sentence, article, document? Long texts produce more matches.
- **Dictionary**: who chose the words, before or after seeing the results?
- **Audit**: is there a reading of a random sample, with the share of errors? By country?
- **Recall**: what provisions use other words?
- **Concept**: text or practice? Is the measure compared with an independent measure of the same concept?
- **Processing**: how were the texts extracted, and with which errors? (One German sentence in our file begins "5. state security.", words left over from the line above.)

Your draft comments

- Check each comment against two lists.

The four steps (Lecture 6)

1. Where: table, column, page.
2. What: a fact that can be checked.
3. Why it matters: which conclusion.
4. What would resolve it.

Berk et al. (Lecture 5)

- Essential or suggestion? Labelled?
- Is the argument scientific, not a hunch?
- Is the request worth its cost?
- Is it about the paper the authors wrote?

DISCUSS Revising a comment

Read one of your comments aloud. The class says:

1. which of the four steps is missing;
2. whether it is essential or a suggestion;
3. one way to make it more specific.

- Then rewrite your weakest comment.

ANSWERS Revising a comment

1. The steps most often missing are 1, where, and 4, what would resolve it. “The identification is weak” names no table and asks for nothing.

2. Essential if a main conclusion of the paper would change; otherwise it is a suggestion and must be labelled as such. A request for a different paper should be dropped.

3. Name the table and column, state a fact that can be checked, and ask for one feasible check or for a narrower claim.

- Tone: write about the paper, not about the authors.

Laboratory

- Worksheet: `exercises/08_text_as_data.do`. Data: `constitutions_text.dta`.
- Tasks:
  1. audit 20 flagged sentences and compute precision;
  2. find what the dictionary misses in Denmark and the Netherlands;
  3. build and audit a second dictionary;
  4. prosecutors instead of judges;
  5. two ways to normalise the count;
  6. one referee comment on a text-based measure.
- Then finish the revision of your comments.

Exit check

1. What are the three steps that turn text into data?
2. Define precision and recall in one sentence each.
3. Give one reason why a constitution can score zero on our measure and still protect the independence of judges.

Next lectures

- Lectures 9–10: your presentations, each followed by questions from the class (format in the syllabus).
- Present the question, one central result, a strength and your most important comment.
- Ask at least one question in each presentation other than your own: your best three questions count.
- Final report: by the common deadline announced on Teams.